

Educational Manual for ISP v1.0

Planning sample size for a two-group human intervention study

Version 1.0 Date: 2026-05-11

Standard problem assumed by this manual

The default problem is a study of an intervention in a human process, such as learning, training, usability, work performance, or comprehension. The manual assumes no medication, surgery, invasive treatment, or high-risk clinical procedure. If the study has clinical, safety, legal, or regulatory consequences, this app can support early planning, but the final design should be reviewed by a qualified statistician and the relevant ethics or regulatory body.

Intervention Sample Planner contributors

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1 What This Manual Is For

This manual explains how to use the **wizard** of ISP v1.0 and how each question changes a parameter in the **configuration tab**. It is written for readers who are not statisticians but need to plan a responsible validation study.

The app helps with one common design:

intervention group vs control group

In this design, one group receives the new intervention and the other does not. The goal is to decide how many valid participants are needed, how to divide them between groups, and how many people to invite or start after losses are considered.

The main idea

Sample size is not only a formula. It is a decision about the strength of evidence needed to support a conclusion. A study should collect enough evidence to answer the question it asks, and it should ask a question that can be answered by the evidence it can realistically collect. This framing follows the sample-size justification perspective described by Lakens (2022).

1.1 What the app can do

ISP v1.0 currently supports:

- two independent groups: **intervention group** and **control group**;
- a **continuous outcome**, analyzed as a comparison between two means;
- a **binary outcome**, analyzed as a comparison between two proportions;
- equal or unequal group allocation;
- two-sided or one-sided hypotheses;
- **alpha**, **power**, and **effect size** planning;
- Bonferroni adjustment for multiple primary comparisons;
- finite-population correction;
- cluster design-effect correction;
- response, completion, usable-data, and buffer corrections;
- a sensitivity table and a narrative justification paragraph.

1.2 What the app cannot cover

The app is intentionally narrower than full statistical software such as G*Power (Faul, Erdfelder, Lang, et al. 2007; Faul, Erdfelder, Buchner, et al. 2020). It does not currently handle:

- repeated-measures or paired pre-post designs;
- non-inferiority or equivalence designs;
- regression, mixed models, mediation, or covariate-adjusted power;
- longitudinal outcomes;
- adaptive designs;
- randomization by cluster with full field-experiment modeling;
- clinical trial design with medication, surgery, safety endpoints, or regulatory requirements;
- qualitative saturation or information power calculations.

If the study is mainly qualitative, use qualitative sample adequacy logic such as information power (Malterud et al. 2016). If the design is quasi-experimental, interrupted time series, matched comparison, or natural experiment, consult causal design guidance such as Shadish et al. (2002). If the intervention is clinical, use dedicated clinical-trial planning references and expert review (Julious 2010).

2 Concepts You Need Before Using the Wizard

2.1 Unit of analysis and unit of observation

The **unit of analysis** is what the conclusion is about. In a learning intervention, it is usually the learner. In a usability intervention, it is usually the user. In a team training intervention, it might be the team.

The **unit of observation** is where the data come from. A learner can produce several observations: a pre-test score, a post-test score, a log file, a questionnaire, and an interview. Many observations from the same person do not automatically become many independent people.

Important limit: Do not count logs as independent people

If 40 people each generate 1,000 log events, the study does not have 40,000 independent participants. The participant-level sample is still 40 unless the analysis explicitly models the dependence between events.

2.2 Outcome type

The **outcome** is the main result used to validate the intervention. ISP v1.0 asks whether it is:

- **continuous**: a score, time, scale, grade, or numeric measurement;
- **binary**: yes/no, success/failure, completed/not completed, returned/did not return, error/no error.

This choice matters because comparing means and comparing proportions use different formulas.

2.3 Alpha, beta, and power

The app uses the classical testing language associated with the Neyman–Pearson framework (Neyman and Pearson 1933).

Alpha is the planned risk of a **Type I error**: concluding that the intervention made a difference when there is no real difference.

Beta is the planned risk of a **Type II error**: missing a real effect that the study was supposed to detect.

Power is:

$$\text{power} = 1 - \beta.$$

In practical language, power is the probability that the study will detect the planned effect if that effect is real.

For readers who want more statistics: Why 0.05 alpha and 0.80 power are common

Values such as $\alpha = 0.05$ and power = 0.80 are conventions, not laws. They are common in behavioral and educational research, but a study can justify more conservative or more permissive values. A higher power target, such as 0.90, usually requires more participants. A smaller alpha, such as 0.01, usually also requires more participants.

2.4 Effect size

The **effect size** is the smallest difference that would matter enough to justify the study.

For a continuous outcome, the app uses **Cohen's d** (Cohen 1988):

$$d = \frac{\mu_{\text{intervention}} - \mu_{\text{control}}}{s}.$$

For a binary outcome, the effect is the difference between two proportions:

$$p_{\text{intervention}} - p_{\text{control}}.$$

Effect size is not a decoration

Do not choose an effect only because it is conventional. Ask what change would matter in the real process. For example, if the intervention is a learning activity, what difference in post-test score would justify adopting it? If the intervention is a tutorial, what increase in completion rate would justify changing the product?

3 How the Wizard Maps to Configuration Variables

The **wizard** asks fewer questions than the **configuration tab** shows. This is intentional. The wizard focuses on the core planning path. The configuration tab contains every editable field, including labels, notes, optional raw means, finite-population correction, and extra buffer.

Wizard question	Configuration variable	Result affected	Why it matters
Study name	<code>study_name</code>	report text	Keeps the calculation traceable.
Outcome type	<code>outcome_type</code>	formula choice	Selects means or proportions.
Cohen's d	<code>effect_size_d</code>	valid sample	Defines the continuous effect to detect.
Control proportion	<code>proportion_control</code>	valid sample	Defines the baseline binary rate.
Intervention proportion	<code>proportion_intervention</code>	valid sample	Defines the expected intervention rate.
Alpha	<code>alpha</code>	<code>alpha_adjusted</code> , <code>z_alpha</code>	Controls false-positive risk.
Power	<code>power</code> , <code>z_power</code>	valid sample	Controls false-negative protection.
Allocation ratio	<code>allocation_ratio</code>	group division	Divides participants between groups.
Response/start rate	<code>response_rate</code>	invited target	Converts assigned target into invitations.
Completion rate	<code>completion_rate</code>	assigned target	Accounts for dropout after start.
Usable data rate	<code>usable_data_rate</code>	assigned target	Accounts for invalid or unusable data.
Primary comparisons	<code>primary_comparisons</code>	<code>alpha_adjusted</code>	Adjusts alpha for multiple planned tests.
Average cluster size	<code>cluster_average_size</code>	<code>design_effect</code>	Accounts for grouped participants.
Intraclass correlation	<code>intraclass_correlation</code>	<code>design_effect</code>	Measures similarity within clusters.

4 Step-by-Step Wizard Guide

4.1 Step 1: Study name

Question in the wizard

What is the study called?

Configuration variable changed

`study_name`

Why this is necessary

The study name appears in the report and saved configuration. It lets you connect the calculation to a protocol, ethics submission, dissertation chapter, or project file.

Applies when

Always.

Does not cover

The name does not define the scientific question. You still need a clear intervention, a control condition, and a primary outcome.

4.2 Step 2: Outcome type

Question in the wizard

What kind of main outcome validates the intervention?

Configuration variable changed

`outcome_type`

Why this is necessary

This is the most important branching question in the wizard. If the outcome is a mean score, the app uses the two-means formula. If the outcome is a rate, the app uses the two-proportions formula.

Real-life examples

- Continuous: post-test score, usability score, engagement scale, time to complete a task.
- Binary: completed the activity, passed the level, returned voluntarily, made an error.

Applies when

Use **continuous** when each participant contributes a numeric score. Use **binary** when each participant contributes a yes/no or success/failure result.

Does not cover

If each participant is measured before and after, a paired or pre-post method may be more efficient. If the outcome is a count, ordinal scale, survival time, or repeated log sequence, use a specialized model.

For readers who want more statistics: Continuous does not mean perfect normality

The two-means approximation is often used for scores and scales, especially when sample sizes are not tiny. However, highly skewed times, bounded scales with floor/ceiling effects, or ordinal data may need different analysis or simulation.

4.3 Step 3A: Cohen's d for continuous outcomes

Question in the wizard

What standardized mean difference would be large enough to matter?

Configuration variable changed

`effect_size_d`

Why this is necessary

For continuous outcomes, the app needs to know the difference it should be able to detect. Cohen's d expresses that difference in standard-deviation units.

$$d = \frac{\text{meaningful mean difference}}{\text{pooled standard deviation}}.$$

Real-life example

Suppose a learning test has a pooled standard deviation of 10 points. If a 5-point improvement is the smallest difference worth adopting the intervention, then:

$$d = 5/10 = 0.5.$$

Configuration alternatives

The wizard asks directly for `effect_size_d`. In the **configuration tab**, you can instead document:

- `mean_control;`
- `mean_intervention;`
- `sd_pooled.`

If `effect_size_d` is blank, the API can compute it from those raw values.

Does not cover

This step does not decide whether the measure is valid. A poorly designed test can produce a precise sample size for the wrong outcome.

4.4 Step 3B: Control and intervention proportions for binary outcomes

Questions in the wizard

What proportion do you expect in the group without intervention? What proportion do you expect in the intervention group?

Configuration variables changed

`proportion_control` and `proportion_intervention`

Why this is necessary

For binary outcomes, the effect is the difference between two rates. Planning to detect a change from 45% to 60% is different from planning to detect a change from 80% to 85%.

Real-life example

In the standard process, 45% of learners complete an activity. With the intervention, the team expects 60% to complete it. The inputs are:

$$p_{\text{control}} = 0.45, \quad p_{\text{intervention}} = 0.60.$$

Does not cover

If the event is rare, if the outcome has several categories, or if participants can experience the event many times, use a method for rare events, multinomial outcomes, or repeated counts.

For readers who want more statistics: Where the two-proportions formula comes from

The app uses a normal approximation to compare two independent proportions. Methods for rates and proportions are a large topic; Fleiss et al. (2003) is a standard reference.

4.5 Step 4: Alpha

Question in the wizard

What alpha should control false positives?

Configuration variable changed

`alpha`

Result variables affected

`alpha_adjusted` and `z_alpha`

Why this is necessary

Alpha controls the planned probability of a false positive under the testing procedure. A common choice is 0.05.

Applies when

Use this when the study will make a statistical comparison between intervention and control.

Does not cover

Alpha does not protect against every form of bias. Non-random assignment, invalid measures, selective reporting, and missing data can still make a result misleading.

4.6 Step 5: Power

Question in the wizard

What power should the study target?

Configuration variable changed

power

Result variable affected

z_power

Why this is necessary

Power controls how likely the study is to detect the planned effect if it really exists. A common value is 0.80; more confirmatory studies often use 0.90 or 0.95.

Does not cover

Power is conditional on the planned effect and assumptions. If the true effect is smaller, the actual chance of detection is lower.

4.7 Step 6: Allocation ratio

Question in the wizard

How will participants be divided between intervention and control?

Configuration variable changed

allocation_ratio

Why this is necessary

The allocation ratio is:

$$k = n_{\text{intervention}}/n_{\text{control}}.$$

Equal groups ($k = 1$) are usually efficient. Unequal groups can be necessary when one group is harder to recruit or when the intervention is available to more people.

Does not cover

This does not solve confounding. If participants choose their own group, differences between groups may reflect pre-existing differences rather than the intervention. Use random assignment when possible, or a quasi-experimental method when randomization is not possible (Shadish et al. 2002).

4.8 Step 7: Response/start rate

Question in the wizard

Of the people invited, what share will start or accept?

Configuration variable changed

response_rate

Result variable affected

invited_needed

Why this is necessary

The calculated sample is valid data, not invitations. If only 40% of invited people start, the number of invitations must be much larger than the number of valid cases.

Example

If 126 valid cases are needed and only 70% of invited people are expected to start, invitations must be inflated:

$$126/0.70 = 180.$$

4.9 Step 8: Completion rate

Question in the wizard

Of the people who start, what share will complete the study?

Configuration variable changed

completion_rate

Result variable affected

assigned_needed

Why this is necessary

Participants may leave after assignment. If the valid target is 63 per group and the expected completion rate is 0.85, the study should start:

$$63/0.85 = 74.12,$$

which rounds to 75 per group.

Does not cover

This correction assumes missingness is a recruitment/planning problem. If dropout is related to the intervention or outcome, the analysis also needs a missing-data plan.

4.10 Step 9: Usable data rate

Question in the wizard

Of the completed cases, what share will provide usable data?

Configuration variable changed

`usable_data_rate`

Why this is necessary

Some completed cases may still be unusable because of failed eligibility criteria, corrupted files, incomplete forms, attention-check failure, or technical errors.

Does not cover

This correction does not justify careless data collection. It is better to improve instruments, instructions, and logging quality than to compensate with a larger sample.

4.11 Step 10: Primary comparisons

Question in the wizard

How many primary comparisons are planned?

Configuration variable changed

`primary_comparisons`

Result variable affected

`alpha_adjusted`

Why this is necessary

Testing many primary outcomes increases the chance of a false positive. The app applies a Bonferroni adjustment:

$$\alpha_{\text{adjusted}} = \alpha/m.$$

The idea is associated with the classical Bonferroni inequality (Bonferroni 1936).

Does not cover

Bonferroni is simple and conservative. Other methods may be better for many correlated outcomes, exploratory analysis, or false discovery rate control.

4.12 Step 11: Cluster average size

Question in the wizard

Are participants clustered in classes, groups, schools, or workshops?

Configuration variable changed

`cluster_average_size`

Why this is necessary

People in the same class or workshop tend to be more similar than independent people. The app uses a design-effect correction:

$$DEFF = 1 + (m - 1)\rho.$$

Applies when

Use this when participants are naturally grouped, such as students in classes, employees in teams, or users in workshop cohorts.

Does not cover

This is a simplified correction. If whole clusters are randomized, if there are few clusters, or if cluster sizes vary strongly, use dedicated cluster-randomized design methods (Donner and Klar 2000).

4.13 Step 12: Intraclass correlation

Question in the wizard

How similar are participants inside the same cluster?

Configuration variable changed

`intraclass_correlation`

Why this is necessary

The intraclass correlation, or **ICC**, measures how much participants inside the same cluster resemble each other. Even a small ICC can substantially increase sample size when clusters are large.

Example

If $m = 25$ and $\rho = 0.05$:

$$DEFF = 1 + (25 - 1)0.05 = 2.2.$$

The independent sample is multiplied by 2.2.

5 The Configuration Tab

The **configuration tab** contains every variable used by the API. Some are asked in the wizard; others are optional or advanced.

Variable	Where used	Meaning
language	interface/report	English or Portuguese display language.
analysis_unit	documentation	Unit about which the conclusion is made.
observation_unit	documentation	Unit from which data are collected.
alternative	formula	Two-sided, intervention greater, or intervention smaller.
mean_control	continuous documentation	Expected raw mean for control.
mean_intervention	continuous documentation	Expected raw mean for intervention.
sd_pooled	continuous effect calculation	Pooled standard deviation.
apply_fpc	finite-population correction	Turns finite-population correction on/off.
finite_population	finite-population correction	Size of the closed eligible population.
extra_buffer_rate	missing-data correction	Extra conservative loss buffer.
intervention_label	report	Name for the intervention group.
control_label	report	Name for the control group.
notes	project memory	Assumptions, citations, pilot evidence, limitations.

Important limit: Finite-population correction

Use `apply_fpc` only when the conclusion is restricted to a known closed population, such as all students in one course. Do not use it for broad populations such as all learners, all players, or all workers.

6 Understanding the Results

The **results** **tab** deliberately separates several sample numbers. This prevents a common planning error: confusing valid data with invitations.

6.1 Initial valid target

`initial_valid` is the theoretical number of valid analyzable cases required if everyone provides usable data.

For a continuous study with $d = 0.5$, $\alpha = 0.05$, power = 0.80, and equal allocation, the app reports:

$$63 + 63 = 126.$$

This is not the number of people to invite. It is the number of valid cases needed for analysis.

6.2 Finite-population adjusted target

`fpc_adjusted_valid` applies only if `apply_fpc` is true. It can reduce the required valid sample when the target population is small and closed.

6.3 Design-adjusted valid target

`design_adjusted_valid` applies the cluster design effect. If clusters are present, this can be much larger than the independent-participant sample.

6.4 Assigned or started target

`assigned_needed` inflates the corrected valid target for completion, usable-data loss, and extra buffer.

6.5 Invited target

`invited_needed` inflates the assigned target for the response/start rate.

6.6 Effective data rate

`effective_data_rate` is the combined rate from invitation to valid data:

$$\text{effective data rate} = \text{response rate} \times \text{completion rate} \times \text{usable data rate} \times (1 - \text{extra buffer}).$$

If this number is low, recruitment risk is high.

6.7 Sensitivity table

The **sensitivity table** asks: what happens if the effect is smaller, the effect is larger, or the desired power changes? This is important because sample-size planning is only as good as its assumptions.

How to read sensitivity

If a small change in effect size creates a very large change in sample size, the design is fragile. A researcher should then consider pilot data, a more precise outcome, a narrower population, or a more modest conclusion.

7 Worked Examples

7.1 Example 1: Teaching intervention with a continuous outcome

Problem

A researcher tests whether a new learning activity improves a post-test score compared with a standard activity.

Inputs

- `outcome_type`: continuous;
- `effect_size_d`: 0.5;
- `alpha`: 0.05;
- `power`: 0.80;
- `allocation_ratio`: 1;
- no losses.

Result

63 control + 63 intervention = 126 valid participants.

Meaning

If the true effect is about $d = 0.5$, the planned study has about 80% power to detect it using the normal approximation. Exact tools may return 64 per group because they use a t-based method (Faul, Erdfelder, Lang, et al. 2007).

7.2 Example 2: Same intervention with 15% dropout

Inputs changed

`completion_rate` becomes 0.85.

Result

$$63/0.85 = 74.12,$$

so the app rounds up to:

75 control + 75 intervention = 150 participants to start.

Meaning

The study still needs 126 valid participants, but it should start 150 people to protect that valid target from expected dropout.

7.3 Example 3: Completion rate as a binary outcome

Problem

The intervention is expected to increase completion from 45% to 60%.

Inputs

- `outcome_type`: binary;
- `proportion_control`: 0.45;
- `proportion_intervention`: 0.60;
- `alpha`: 0.05;
- `power`: 0.80;
- `allocation_ratio`: 1.

Result

$$173 \text{ control} + 173 \text{ intervention} = 346 \text{ valid participants.}$$

Meaning

Rates often need larger samples than scores. A difference of 15 percentage points can be important in practice, but it still requires many observations to estimate reliably.

7.4 Example 4: Classroom clusters

Problem

The same continuous intervention is delivered in classes of about 25 learners. Learners inside the same class are more similar than independent learners.

Inputs changed

- `cluster_average_size`: 25;
- `intraclass_correlation`: 0.05.

Design effect

$$DEFF = 1 + (25 - 1)0.05 = 2.2.$$

Result

$$126 \times 2.2 = 277.2,$$

rounded to 278 valid participants:

$$139 \text{ control} + 139 \text{ intervention.}$$

Meaning

Clustered data can greatly increase the required sample. Adding more classes may help more than adding more students inside the same class.

8 When to Use Another Methodology

Use another method when the research question is not a two-group independent comparison.

Situation	Why this app is not enough	Consider instead
Same person measured before and after	Measurements are paired, not independent.	Paired t-test or repeated-measures planning.
Whole classes randomized	Number of clusters matters strongly.	Cluster-randomized trial methods.
No control group	Causal claim is weak.	Pilot, pre-post, interrupted time series, or quasi-experimental design.
Main evidence is interview data	Power formulas do not apply.	Qualitative information power or saturation logic.
Outcome is time-to-event	Censoring and timing matter.	Survival analysis design.
Outcome is count data	Variance structure differs.	Poisson or negative-binomial planning.
Clinical safety or medication	Regulatory and ethical stakes are higher.	Clinical-trial statistical planning.

9 Glossary of Important Terms

Alpha Planned Type I error rate.

Allocation ratio Planned size of intervention group divided by control group.

Binary outcome A yes/no or success/failure outcome.

Cluster A natural group such as class, team, school, or workshop.

Cohen's d Standardized mean difference.

Completion rate Share of starters expected to finish.

Control group Group without the intervention.

Design effect Multiplier that adjusts for clustered data.

Effect size Smallest difference that matters for the study.

Finite population Closed known population to which the conclusion is restricted.

ICC Intraclass correlation.

Intervention group Group receiving the intervention.

Power Probability of detecting the planned effect if it exists.

Response rate Share of invited people expected to start or accept.

Usable data rate Share of completed cases expected to be analyzable.

Valid sample Participants with usable analyzable data.

Wizard Guided interface mode that asks core planning questions.

10 Final Practical Checklist

Before collecting data, write down:

1. the intervention;
2. the control condition;
3. the target population or context;
4. the analysis unit and observation unit;
5. the primary outcome;
6. whether the primary outcome is continuous or binary;
7. the smallest effect that matters;
8. alpha, power, and hypothesis direction;
9. allocation ratio;
10. expected response, completion, and usable-data rates;
11. whether finite-population or cluster correction applies;
12. the final valid target, assigned target, and invited target;
13. the limits of the conclusion.

A good conclusion is proportional

The safest final sentence is often not "the intervention works." It is: "Under these conditions, with this outcome, this comparison, and this amount of valid data, the study was planned to detect an effect of this size with this level of uncertainty."

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